How Much Context Does LM Surprisal Need to Predict Reading Times? Evidence from Turkish Eye Tracking

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Abstract

We examine how much of the preceding text a language model (LM) needs to predict human reading times, and whether this depends on model size. Using the eye movements of 57 Turkish readers, six LMs from three families, and held-out comparisons of mixed-effects models, we compared surprisal from three contexts: the preceding word (2-gram), the current sentence (sentential), and the full passage (full). Sentential surprisal predicted first fixation duration, gaze duration and total duration significantly better than 2-gram surprisal, and full-context surprisal did not significantly improve on sentential surprisal in any measure. The largest full discourse context improvement was in total duration, though non-significant, and it was larger for readers with higher comprehension accuracy. Smaller models fit reading times slightly better than larger ones within each family, in the direction of inverse scaling, but these differences varied between passages and were not significant once this variability was modelled. These findings indicate that the predictive value of LM surprisal for Turkish reading times comes mainly from the current sentence, and that surprisal conditioned on earlier sentences may add predictive value mainly for a late measure and for readers with higher comprehension accuracy.

1 Introduction

In this study, we examine how much of the preceding text a language model (LM) needs to predict human reading times, and whether this depends on the size of the LM. Words that are less predictable in their context are read more slowly (Demberg and Keller, 2008; Smith and Levy, 2013). A common measure of predictability is surprisal, the negative log probability of a word given the preceding words (Hale, 2001; Levy, 2008). Surprisal has been estimated with probabilistic grammars (Hale, 2001) and n-gram models (Smith and Levy, 2013), and it is now usually estimated with LMs. LM surprisal predicts reading times in many languages (Wilcox et al., 2023; de Varda and Marelli, 2023).

In surprisal theory, the probability of a word is conditioned on the preceding words of the sentence and, in principle, also on context outside the sentence (Levy, 2008). However, the theory was developed and tested mainly on sentence-level phenomena, and early estimates of surprisal relied mainly on the current sentence or on the last few words (Hale, 2001; Smith and Levy, 2013). The theory therefore does not predict how much earlier sentences contribute to the predictability of a word. Research on human sentence processing does not give a single answer either. Some studies found that discourse context did not affect the initial parse (Ferreira and Clifton, 1986), while others found early effects of discourse context on reading times (Altmann and Steedman, 1988) and evidence that readers predict specific words from the discourse (Van Berkum et al., 2005).

Studies that limit the context of an LM have addressed a related question. Surprisal from LMs that saw only the preceding word predicted English and Japanese gaze durations better than surprisal from LMs that saw the whole sentence (Kuribayashi et al., 2022). When the context could extend across sentences, English reading times were fit well by very short windows (fewer than 20 tokens) and best by long windows of 500 to 1,000 tokens (Kajikawa et al., 2026). These studies compare a short window with a long one. A long window, however, contains two kinds of information: the earlier words of the current sentence (sentential context) and the earlier sentences of the text (discourse context). Such comparisons do not show which of the two improves the prediction of reading times.

We separate the two kinds of information in Turkish, an agglutinative, head-final language with canonical SOV order that has been studied less in reading research (Acartürk et al., 2024). We use

the eye movements of 57 readers who read 10 passages from four genres in the Turkish part of the MultiplEYE corpus (Jakobi et al., 2025), and six LMs from three families that differ in size. For every word, we compute surprisal from three contexts of increasing size: the preceding word only (2-gram context), the preceding words of the current sentence (sentential context), and all preceding words of the passage (full context). The surprisal estimates from each context are entered into separate mixed-effects models of reading times, and these models are compared by how well they predict reading times that were not used to fit them (following Wilcox et al., 2023).

We ask three questions. First, does context beyond the preceding word improve the prediction of reading times, and how much of this improvement comes from the current sentence? Second, do earlier sentences improve the prediction further, and does this differ between early eye-movement measures, which reflect first-pass processing, and later measures, which include re-reading? Third, does model size change how much the LMs depend on context? Larger LMs can predict reading times worse (Oh and Schuler, 2023), and they also use context differently: they reach a lower loss with less preceding context and continue to improve as the context grows (Kaplan et al., 2020). Model size may therefore change how much sentential and discourse context contribute to the prediction of reading times. Throughout, we study LM surprisal as a predictor of human reading times; we do not claim that the context given to an LM corresponds to what readers keep in memory.

We found that sentential context predicted first fixation duration, gaze duration and total duration significantly better than the 2-gram context, and full context did not significantly improve the prediction beyond sentential context in any measure. The full–sentential difference was largest in total duration, and it was larger for readers with higher comprehension accuracy. Model size did not significantly change these results.

2 Background

Context in LM-based models of reading. Several studies have asked how much context LM surprisal should use (Kuribayashi et al., 2022; Michaelov and Levy, 2026; Kajikawa et al., 2026). Kuribayashi et al. (2022) limited the input of LMs to 2- to 20-gram contexts (the preceding 1 to 19 words) of the current sentence or to the whole sentence, and found that the shortest window, the preceding word alone, gave the best fit to English and Japanese gaze durations. Kajikawa et al. (2026) varied the input of GPT-2 from fewer than 20 to about 1,000 tokens: very short windows fit reading times well, and windows of 500 to 1,000 tokens fit best. When coreference chains across sentences were broken, the advantage of the long windows decreased by 20 to 40%. Neither study compared the current sentence with the full preceding text. We make this comparison in the present study.

Discourse context in human reading. Discourse context can affect reading in at least two ways. It can affect structural decisions, such as which attachment of an ambiguous phrase is chosen. Whether this happens during the initial parse is debated: Ferreira and Clifton (1986) found that a biasing discourse context did not remove the first-pass reading difficulty of nonminimal attachments, whereas Altmann and Steedman (1988) found that referential context affected reading times at the disambiguating region. Discourse context can also make specific words predictable: readers anticipate upcoming words from the preceding discourse (Van Berkum et al., 2005). Surprisal concerns the latter. Eye tracking is useful for this question because it separates measures of first-pass reading from measures that include re-reading (Clifton et al., 2007), so it can show at which stage a contribution of discourse context appears.

Model size. The inverse relation between model size and fit to reading times (Oh and Schuler, 2023; Oh et al., 2025; Oh and Linzen, 2026) differs between early and late eye-movement measures across languages (de Varda and Marelli, 2023). One explanation is that larger models assign very low surprisal to infrequent words such as names, which readers still find difficult (Oh and Schuler, 2023). In Turkish self-paced reading, smaller GPT-2 models predicted reading times better than a larger LLaMA model (Keleş and Deniz, 2025). Larger LMs also use context differently: they reach a lower loss with less preceding context and continue to improve as the context grows (Kaplan et al., 2020). Whether larger and smaller models therefore differ in how much context they need to predict reading times has not been tested.

Differences between readers. Readers differ in how much effort they make to connect the current

text to what they have read before (van den Broek et al., 2011), and re-reading supports comprehension (Schotter et al., 2014). If earlier sentences contribute to reading, their contribution may therefore be larger for readers who understand the text better. We test this by relating the contribution of discourse context to comprehension accuracy.

3 Methods

3.1 Eye-tracking data

Participants. Fifty-seven undergraduate students who were native speakers of Turkish took part in the Turkish data collection of MultiplEYE (Jakobi et al., 2025) (47 women, 10 men; mean age 22.3 years, $SD = 4.9$, range 18–41; mean years of education 15.2). Two participants were bilingual speakers (of Arabic and Turkish, or of Kurmanji and Turkish). The study was approved by the ethics committee of the authors' institution. Participants gave informed consent and received course credit in return for their participation.

Materials. Participants read 10 passages from four genres (Table 1), taken from the MultiplEYE text corpus (Kasperė et al., 2026). The passages ranged from 252 to 719 words (4,702 words and 305 sentences in total). Their sources are listed in Appendix A. Each passage was shown over several screens (5 to 14 per passage) in a monospaced font (JetBrains Mono, 24 px). Participants moved to the next screen with a key press and could not return to earlier screens. After each passage, they rated their familiarity with the text and its difficulty and answered six comprehension questions with three options each. Two questions required local inference, two required bridging inference and two required global inference. Mean accuracy was 66.3% (chance: 33%).

Apparatus and procedure. Eye movements were recorded with an EyeLink 1000 Plus (SR Research, tower mount) at 1000 Hz. Stimuli were shown on a 24-inch monitor (53 × 30 cm, 1920 × 1080 px) at a viewing distance of 60 cm. The session followed the MultiplEYE protocol (Figure 4 in Appendix A): consent, instructions, calibration, two practice texts, and the 10 passages in pseudo-randomized order.

Reading measures. Fixations were mapped to words with the MultiplEYE preprocessing pipeline. We analysed four measures (Appendix A). First fixation duration (FFD) and gaze duration (GD), the sum of all first-pass fixations on a word, reflect early processing. Total duration (TD), the sum of all fixations on a word, and re-reading duration (RRD), the fixation time on a word after the first pass, include later processing. Of the 268,014 participant-word pairs, 37.2% were skipped in the first pass and 30.0% were never fixated. We excluded zero durations, durations above the mean plus three standard deviations, observations whose current or previous word consisted only of punctuation, and the first two words of each screen.

3.2 Language models and surprisal

We used six base autoregressive LMs from three families (Table 2). Kanarya-750m and Kanarya-2b were trained on Turkish web text (Safaya et al., 2022), cosmosGPT small, medium and large are GPT-2 models trained on Turkish data (Kesgin et al., 2024), and mGPT is multilingual (Shliazhko et al., 2024).

The surprisal of a word w_i given its context c_i is $S(w_i) = -\log P(w_i \mid c_i)$. For a word that consists of the subword tokens $t_1, \ldots, t_k$, we summed the token surprisals:

$$S(w_i) = \sum_{j=1}^{k} -\log P(t_j \mid c_i, t_1, \ldots, t_{j-1}). \quad (1)$$

Surprisal was computed on the running passage and measured in nats. A token was assigned to the word that contains its last character, so a token with a leading space belongs to the following word. We varied only the context c_i:

- **2-gram context**: the preceding word only, so that the LM sees the same two words as a 2-gram model (as in Kuribayashi et al., 2022). To show how fit changes with context length, we also used n-gram contexts with n from 3 to 200, that is, the preceding 2 to 199 words (Appendix D).
- **Sentential context**: the preceding words of the current sentence.
- **Full context**: all preceding words of the passage, including words on earlier screens, up to the token limit of the LM. Full context contains both the sentential context and the earlier sentences of the passage.

The difference between full and sentential context is therefore the contribution of the earlier sentences

Passage	Genre	Scr.	Words	Sent.	W/S	Tokens per word			Acc.
						Kan.	cos.	mGPT	
Cow's milk (PISA)	Argumentative	14	716	59	12.1	1.58	1.49	2.43	72.8
Rapa Nui (PISA)	Argumentative	13	496	43	11.5	1.69	1.58	2.46	66.7
Human Rights (preamble)	Institutional	8	252	2	126.0	1.37	1.30	2.29	79.2
Learning mobility report	Institutional	9	437	21	20.8	1.45	1.37	2.37	59.4
The Alchemist	Literary	5	327	31	10.5	1.56	1.47	2.46	75.7
Broken April	Literary	10	389	29	13.4	1.66	1.57	2.49	62.0
The Magic Mountain	Literary	6	388	19	20.4	1.43	1.37	2.26	59.4
Solaris	Literary	14	719	44	16.3	1.63	1.53	2.44	54.1
Caveman	Popular science	8	318	18	17.7	1.60	1.52	2.44	63.5
The MultiplEYE project	Popular science	11	660	39	16.9	1.43	1.35	2.30	69.9
Total		98	4,702	305	15.4				66.3

Table 1: The 10 passages. Scr.: screens; Sent.: sentences; W/S: words per sentence; tokens per word: subword tokens per word for the three tokenizers; Acc.: mean comprehension accuracy (% correct). The Human Rights passage consists of two long sentences.

Model	Params	Layers	Context	Training data
Kanarya-750m	750M	12	2048	Turkish
Kanarya-2b	2B	24	2048	Turkish
cosmosGPT-S	0.1B	12	1024	Turkish
cosmosGPT-M	0.4B	24	1024	Turkish
cosmosGPT-L	0.8B	36	1024	Turkish
mGPT	1.3B	24	512	61 languages

Table 2: Language models. Context: maximum number of tokens used (Appendix B).

(discourse context). Surprisal was highest with the 2-gram context: for 96% of the analysed words, 2-gram surprisal was higher than full-context surprisal (median difference 4.0 nats). Sentential and full-context surprisal were close in that they differed by less than 1 nat for 68% of the words (median difference 0.56 nats), and they were highly correlated ($r = .86$). Figure 5 in Appendix B shows the three estimates for one sentence.

3.3 Regression models

For each LM, context and reading measure, we fitted a linear mixed-effects model of the log reading time of participant r on word w_i of passage p:

$$\log RT_{rpi} = \boldsymbol{\beta}^\top \mathbf{x}_i + b_r + u_p + \sum_{k=0}^{2} \theta_k S(w_{i-k}) + \varepsilon_{rpi}. \quad (2)$$

The baseline predictors $\mathbf{x}_i$ were the length (in characters) and the Zipf frequency (van Heuven et al., 2014) of w_i and w_{i-1}, taken from wordfreq (Speer, 2022), their length $\times$ frequency interactions, the log position of the word in the passage, and binary predictors for line-initial and line-final words and for sentence-final and clause-final punctuation. b_r and u_p are random intercepts for participants and passages. Reading time on a word is also affected by the predictability of the preceding words (spillover). The model therefore included the surprisal of the target word and of the two preceding words, each standardized with the mean and standard deviation of the training data. All models were fitted by maximum likelihood with `lme4` (Bates et al., 2015) on the same observations, namely the words for which surprisal was available in all contexts for all six LMs. The first word of a sentence has no sentential context, and the model needs the surprisal of the two preceding words. The analysed words therefore started at the fourth word of each sentence, and the 2-gram context of these words never crossed a sentence boundary.

3.4 Model comparison

We compared the reading-time models by their log-likelihood on data not used to fit them, so that models with more predictors were not favored (cf. Wilcox et al., 2023). The unit of cross-validation was a block, the observations of one participant on one passage. Blocks were divided into five folds, the same folds for all LMs and contexts. A held-out block combined a participant and a passage that both appeared in the training folds, so its predictions included their estimated random intercepts. Following the order of the contexts, we made two comparisons: sentential with 2-gram context, and full with sentential context. For each block and each comparison of a larger context a with a smaller context b, we computed the mean difference in held-out log-likelihood per 1,000 ob-

servations:

$$\Delta\mathrm{LL}_{rp} = \frac{1000}{N_{rp}} \sum_{i \in (r,p)} \left[\log p_a(y_i) - \log p_b(y_i)\right]. \quad (3)$$

Positive values mean that surprisal from the larger context predicted reading times better. We averaged $\Delta\mathrm{LL}_{rp}$ over the six LMs, so each block contributed one value per comparison. Results for each LM are given in Appendix D.

Statistical inference. The block-level differences were analysed with a second mixed-effects model, $\Delta\mathrm{LL}_{rp} = \mu + a_r + v_p + e_{rp}$, with random intercepts a_r for participants and v_p for passages, so that both participants and passages were treated as random factors (Clark, 1973; Baayen et al., 2008). We tested whether the mean difference μ was zero with a likelihood-ratio test (LRT) against a model without μ and with the same random effects. The p-values were Holm-corrected over the eight main comparisons (two comparisons by four measures; p_H). We considered a comparison significant if $p_H < .05$. We also report 95% confidence intervals and Holm-corrected p-values from Satterthwaite tests ($p_{S,H}$), which are more conservative, in Table 3.

3.5 Memorization of the stimuli by the LMs

For each passage and LM, we gave the model 12 prompts from the passage (three start positions, each with prefixes of 5, 10, 25 and 50 words) and generated a greedy continuation (Carlini et al., 2023). We recorded the longest run of words that matched the original text and the mean full-context surprisal of the passage. The two Kanarya models reproduced the Human Rights passage (§4.2), so their surprisal for this passage reflects memory of the document rather than predictability from the preceding text. To keep the same data for all LMs, the main analyses therefore excluded this passage for all LMs, and all first-stage models were refitted on the remaining nine passages. Results with all 10 passages are reported as a check. Figure 2 shows the memorized passage.

3.6 Model size

We compared models of different sizes within each family: Kanarya-750m with Kanarya-2b, and the mean of cosmosGPT-S and -M with cosmosGPT-L. First, we compared their fit with full context, using the same block-level ΔLL comparison; positive values favour the smaller models. Second, we compared how much fit each model lost when full context was replaced by the 2-gram context; positive values mean that the larger model lost less.

3.7 Comprehension accuracy

We asked whether the contribution of discourse context differed between readers with higher and lower comprehension accuracy, the proportion of correct answers to the 60 questions on all 10 passages. For each participant-passage block, the outcome was the reduction in squared prediction error when full-context surprisal replaced sentential-context surprisal (mean of the six LMs):

$$\Delta\mathrm{MSE}_{rp} = 1000 \cdot \overline{(y - \hat{y}_{\mathrm{sentential}})^2 - (y - \hat{y}_{\mathrm{full}})^2}. \quad (4)$$

Positive values favour full context. We regressed ΔMSE on standardized accuracy and on the participant's mean log reading time (reading speed), with a random intercept for participants and a random intercept and accuracy slope for passages, and tested accuracy with an LRT. TD was the main measure and GD was analysed for comparison (Holm correction over the two). As a check, we also compared participants above and below the median accuracy. Eight participants scored exactly at the median, so we report this split with these participants assigned to the lower and to the higher group. Finally, we tested whether the association with accuracy differed between genres (accuracy $\times$ genre) and between local, bridging and global questions (a model with the three accuracies, compared with a model in which their slopes were equal).

4 Results

4.1 Sentential and full context

Table 3 shows both comparisons for each measure. Sentential surprisal predicted reading times significantly better than 2-gram surprisal in FFD (ΔLL = 0.90, 95% CI [0.37, 1.42], $p_H = .013$), GD (1.76 [0.86, 2.66], $p_H = .006$) and TD (2.49 [0.83, 4.16], $p_H = .026$). In RRD, the difference was not significant (0.96 [0.04, 1.87], $p_H = .137$). The point estimate of the sentential–2-gram difference was positive for each of the six LMs in all four measures (Table 10 in Appendix D). With all 10 passages, the results were the same except that the TD difference was not significant after correction (2.11 [0.43, 3.79], $p_H = .070$).

The differences between full and sentential context were smaller, and none was significant after correction. The largest difference was in TD (1.64

[0.05, 3.23], $p_H = .137$). Before correction, TD was the only measure with $p < .05$ in both tests (LRT $p = .029$, Satterthwaite $p = .044$). The differences in FFD (0.43, $p_H = .226$), GD (0.35, $p_H = .644$) and RRD (0.24, $p_H = .644$) were smaller. The point estimate of the full–sentential difference was also positive for each of the six LMs in all four measures (Table 10). With all 10 passages, no full–sentential difference was significant either (TD: 1.50 [0.10, 2.90], $p_H = .130$).

Figure 1 compares full context with n-gram contexts of increasing length. The advantage of full context became smaller as the n-gram context became longer. After correction over the ten contexts, full context predicted reading times significantly better than the 2-gram context in GD ($p_H = .039$) and TD ($p_H = .002$), and than the 3-gram context in TD ($p_H = .008$; Appendix D). None of the n-gram contexts predicted reading times significantly better than full context. Thus, we did not find the advantage of very short contexts that was reported for English and Japanese (Kuribayashi et al., 2022).

4.2 Memorization of one passage

The two Kanarya models had memorized the Human Rights passage, a public document (Appendix C). Kanarya-2b assigned it a mean surprisal of 0.37 nats per word, compared with 3.9 to 6.7 nats for the other passages, and continued one prompt with 36 words that matched the original exactly. Kanarya-750m showed the same pattern in a weaker form, and the other LMs did not. Most readers did not know the text: only 6 of the 57 participants had read it before. For the two Kanarya models, full context therefore assigned very low surprisal to the words of this passage, although readers did not know the text. As a result, on this passage, full context predicted TD clearly worse than the 2-gram context for these two models (ΔLL = -5.26 and -5.55, compared with means of 3.56 and 4.27 for the same models on the other nine passages; Figure 2). For this reason, the main analyses excluded this passage (§3.5).

4.3 Model size

Model size did not significantly change the fit or the dependence on context (Table 7 in Appendix D). With full context, the smaller models predicted reading times slightly better in both families, in the direction of inverse scaling (Oh and Schuler, 2023): by 0.43 [-0.20, 1.06] (GD) and 0.55 [-0.35, 1.46] (TD) for Kanarya, and by 0.17 and 0.19 for cosmosGPT. When full context was replaced by the 2-gram context, the larger models lost slightly less fit than the smaller models (differences of 0.26 to 0.72). None of these differences was significant ($p_H \geq .40$ without the Human Rights passage; $p_H \geq .24$ with all 10 passages).

The size differences varied strongly between passages. In the Kanarya family, the smaller model predicted TD better in seven of the nine passages (passage means from -1.3 to 3.1). Without a random intercept for passages, the Kanarya differences would be significant ($p < .001$); with it, the estimates were unchanged but their standard errors were 2.4 (GD) and 2.7 (TD) times larger.

4.4 Comprehension accuracy

In TD, the benefit of full over sentential context was larger for participants with higher comprehension accuracy (Figure 3). With reading speed controlled, an increase of one standard deviation in accuracy was associated with an increase of 0.44 [0.11, 0.77] in ΔMSE ($p_H = .016$). The association was positive for each of the six LMs (Appendix D) and was also significant with all 10 passages (0.39 [0.08, 0.70], $p_H = .021$). In GD, the association was small and not significant (0.11, $p_H = .095$).

Splitting participants at the median accuracy also showed a larger full–sentential benefit in the higher-accuracy group. The benefit in this group was significant when the eight participants at the median were assigned to it ($p_H = .026$) but not otherwise ($p_H = .079$). The association between accuracy and the benefit did not differ significantly between genres ($\chi^2(3) = 1.25$, $p = .74$) or between question types ($\chi^2(2) = 1.66$, $p = .44$; Appendix D). These results are correlational: comprehension and reading behaviour may have common causes, and we cannot tell from these data why better comprehenders showed a larger benefit.

5 General Discussion

Sentential and discourse context. Surprisal from the current sentence predicted the reading times of Turkish readers better than surprisal from the preceding word alone, in first fixation duration, gaze duration and total duration. Earlier sentences did not significantly improve the prediction in any measure after correction.

Early and late measures. In first fixation duration and gaze duration, the current sentence improved the prediction significantly and the earlier

Comparison	Measure	Without HR (main) ΔLL [95% CI]	p_H	$p_{S,H}$	All 9 passages ΔLL [95% CI]	p_H
Sentential − 2-gram	FFD	0.90 [0.37, 1.42]	.013	.025	0.83 [0.35, 1.32]	.014
Sentential − 2-gram	GD	1.76 [0.86, 2.66]	.006	.012	1.52 [0.57, 2.46]	.019
Sentential − 2-gram	TD	2.49 [0.83, 4.16]	.026	.048	2.11 [0.43, 3.79]	.070
Sentential − 2-gram	RRD	0.96 [0.04, 1.87]	.137	.215	0.75 [−0.17, 1.66]	.231
Full − sentential	FFD	0.43 [−0.11, 0.98]	.226	.311	0.44 [−0.03, 0.91]	.185
Full − sentential	GD	0.35 [−0.84, 1.54]	.644	.645	0.39 [−0.65, 1.42]	.663
Full − sentential	TD	1.64 [0.05, 3.23]	.137	.215	1.50 [0.10, 2.90]	.130
Full − sentential	RRD	0.24 [−0.24, 0.73]	.644	.645	0.21 [−0.21, 0.63]	.663

Table 3: Sentential context compared with the 2-gram context, and full context compared with sentential context (mean of six LMs). ΔLL: difference in held-out log-likelihood per 1,000 observations, with 95% Satterthwaite intervals, not corrected for multiple comparisons; positive values favour the larger context. p_H: LRT, Holm-corrected over the eight comparisons within each sample; $p_{S,H}$: Satterthwaite test, Holm-corrected. HR: Human Rights passage, excluded from the main analyses (§3.5).

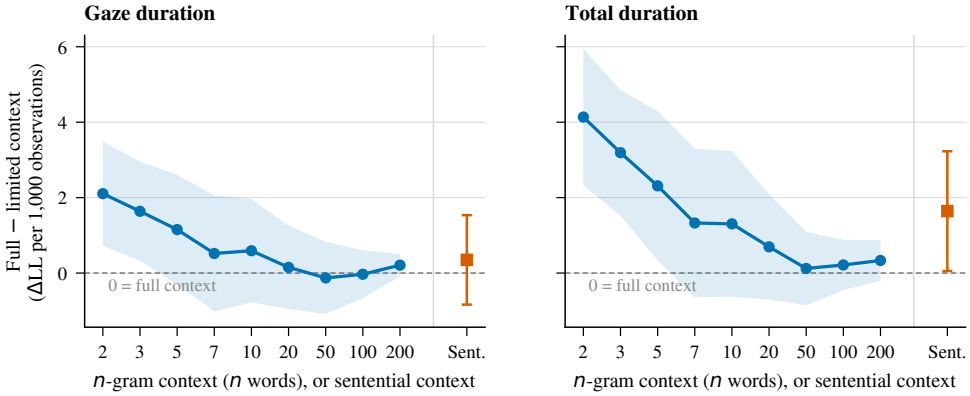


Figure 1: Full context compared with n-gram contexts (n = 2 to 200) and with sentential context (labelled 'Sent.') (mean of six LMs, nine passages, 95% Satterthwaite intervals). Values above zero mean that full context predicted reading times better.

sentences added little (0.43 and 0.35). In total duration, which includes re-reading, the current sentence also improved the prediction significantly (2.49), and the earlier sentences gave the largest, non-significant further improvement (1.64), which was larger for readers with higher comprehension accuracy. In re-reading duration, neither step was significant. This pattern would be in line with proposals that discourse information is used mainly after the initial parse (Ferreira and Clifton, 1986), although that work concerned structural decisions rather than word predictability. However, the measures were not compared statistically and the total-duration difference was not significant, so our data do not provide evidence for this view.

Why discourse context may contribute little. Our results are compatible with two accounts of human sentence processing in which earlier sentences contribute little to the predictability of individual words. In cue-based retrieval accounts, only a limited number of items is held in an active state, and earlier material is accessed through retrieval cues provided by the current input (Lewis and Vasishth, 2005; Lewis et al., 2006). Pronoun antecedents, for example, appear to be retrieved rather than maintained in an active state (Foraker and McElree, 2007). On this account, earlier sentences should affect expectations mainly when the current word provides a cue to them. Lossy-context surprisal makes a related prediction for a different reason: information further back in the context is less available to the comprehender (Futrell et al., 2020). Both accounts predict a small average contribution of earlier sentences. Models of text comprehension describe a similar process at the discourse level. In the RI-Val model, earlier text is reactivated through overlap with the current input and is integrated and validated in parallel (Cook and O'Brien, 2014), and how far this proceeds before the reader moves on depends on a coherence threshold (O'Brien and Cook, 2016). Readers also differ

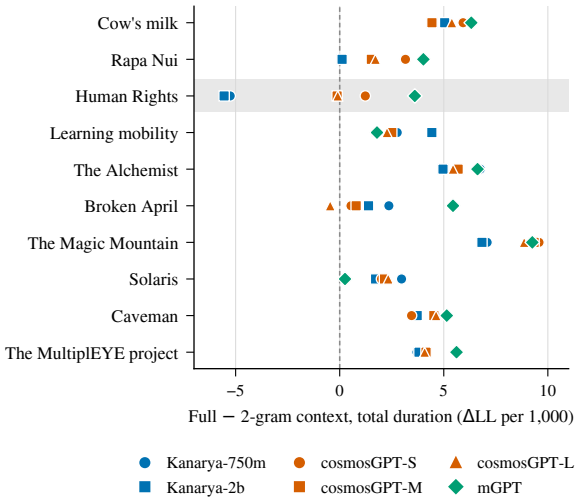


Figure 2: Full context compared with the 2-gram context for total duration, by passage and LM (mean over participants; all 10 passages). On the Human Rights passage (shaded), full context predicted reading times clearly worse than the 2-gram context only for the two Kanarya models, which had memorized this passage.

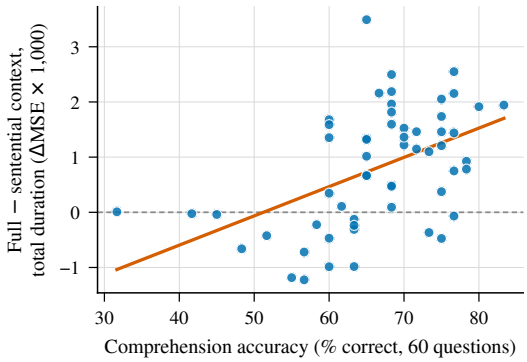


Figure 3: Comprehension accuracy and the benefit of full over sentential context in total duration, one point per participant (mean over nine passages and the six LMs), with a least-squares line. The test in the text uses the mixed-effects model of §3.7.

in the standards of coherence they apply (van den Broek et al., 2011), which would be consistent with the association between comprehension accuracy and the benefit of full context. Our method does not test these mechanisms: restricting the context of an LM changes the predictor, not the reader's memory, and a similar pattern of fit does not show that LMs use context in the same way as human readers.

Comparison with previous LM studies. Unlike Kuribayashi et al. (2022), we found no advantage of very short contexts, in line with the good fit of long windows in English (Kajikawa et al., 2026). Because Japanese and Turkish are both head-final and agglutinative, a typological explanation seems less likely. One methodological difference is that their Japanese LMs were trained on short text segments, whereas our LMs were trained on full documents and may give less reliable estimates from very short contexts.

Model size and memorization. Across the nine passages, the Kanarya size difference was larger in passages where the Kanarya tokenizer split words into more subword tokens ($r = .75$ for TD, .80 for GD). It was largest for the Rapa Nui passage, which contains many names and dates, and it was negative only for the Learning Mobility report and the MultiplEYE description, which have among the fewest tokens per word. This is in line with the account that larger models underestimate the difficulty of rare and complex words (Oh and Schuler, 2023, 2025). Comparisons based on a few texts should treat texts as a random factor and check the stimuli for memorization.

6 Conclusion

Previous studies that limited the context of an LM compared short and long windows without separating the current sentence from the earlier text. When we separated the two in Turkish eye-tracking data, the results showed that surprisal from the current sentence predicted first fixation duration, gaze duration and total duration significantly better than surprisal from the preceding word alone. Earlier sentences did not significantly improve the prediction in any measure. Their largest, non-significant contribution was in total duration, and in an exploratory analysis it was larger for readers with higher comprehension accuracy. Because the estimates varied considerably between texts and the full–sentential comparison had little power with nine passages, testing whether surprisal conditioned on earlier sentences improves the prediction of late measures, and for which readers, will require more texts. A comparison with an equally long but unrelated context would also be needed to separate the relevance of earlier text from its amount.

Limitations

Our data come from one language, one corpus and 10 passages, of which nine were used in the main analyses, so variability between passages is estimated with limited precision, and genre cannot be

separated from passage. The comparison of sentential and full context has little power for small differences, and a non-significant difference is not evidence that the two are equivalent. Full context differs from sentential context both in the amount of text and in its relevance; a comparison with an equally long but unrelated context, and predictors for words repeated from earlier sentences, would be needed to attribute a difference to discourse information. The LM families also differ in tokenization, which can affect the fit of surprisal (Oh and Schuler, 2025). Because Turkish words are often split into several subword tokens, differences between families may partly reflect how words are segmented rather than how much context the models use. In addition, all models have at most 2B parameters. The model-size comparisons therefore cover a limited range, and it remains open whether larger LMs, which can fit reading times differently, would depend on sentential and discourse context in the same way.

Ethics Statement

The eye-tracking data were collected under the MultiplEYE protocol with approval from the ethics committee of the authors' institution; participants gave informed consent and received course credit. The passage that one model family reproduced is a public document. We do not reproduce the copyrighted literary passages. The sentence in Figure 5 is from the MultiplEYE project's own description. Analysis code will be released with the paper.

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A Passages and Reading Measures

Table 4 lists the source of each passage. The Turkish texts, screen layouts and comprehension questions are part of the MultiplEYE stimulus set and are available from the MultiplEYE project; the original sources follow Kasperė et al. (2026). Table 5 defines the four reading measures, and Figure 4 shows the procedure of a session.

ID	Genre	Source and section shown
1	Popular science	Description of the MultiplEYE COST Action, written by Action members
2	Institutional	Universal Declaration of Human Rights (1948), Turkish text; preamble
3	Institutional	European Commission (2017), progress report on the learning mobility benchmark; opening sections
4	Literary	Paulo Coelho, *The Alchemist* (*Simyacı*); Part 1, opening
6	Literary	Thomas Mann, *The Magic Mountain* (*Büyülü Dağ*); foreword
8	Literary	Stanisław Lem, *Solaris*; Chapter 2, opening
9	Literary	Ismail Kadare, *Broken April* (*Kırık Nisan*); Chapter III, opening
10	Argumentative	PISA 2018 reading unit "Cow's Milk" (OECD)
11	Argumentative	PISA 2018 reading unit "Rapa Nui" (OECD)
12	Popular science	Ian Sample (2014), "Swarthy, blue-eyed caveman revealed using DNA from ancient tooth", *The Guardian*

Table 4: Sources of the passages. ID: MultiplEYE stimulus ID; IDs 7 and 13 are the practice texts.

	Definition	N	Mean
FFD	Duration of the first fixation on the word	134,505	198
GD	Sum of first-pass fixations before the eyes leave the word	133,686	233
TD	Sum of all fixations on the word	148,010	304
RRD	Fixation time on the word after first pass	47,515	261

Table 5: Reading measures. N: participant-word observations with all 10 passages (without Human Rights: 125,924, 125,199, 138,583 and 44,099). Mean in ms.

B Surprisal and Model Fitting

Surprisal was computed with the Hugging Face transformers library (version 4.57.6) in 32-bit precision, with model revisions fixed. Word surprisal was the sum of the surprisals of the tokens that belonged to the word, including attached punctuation. We did not apply the correction proposed by Pimentel and Meister (2024), which adjusts word probabilities for tokenizers that mark the beginning rather than the end of words. Kanarya and mGPT received a beginning-of-text token and cosmosGPT did not, based on losses with and without the token. mGPT was trained with sequences of 512 tokens (Shliazhko et al., 2024). Without a limit, its full-context surprisal rises from about 5.4 nats per word within the first 512 tokens to 9.5 to 16 nats beyond them, so we limited its context to 512 tokens. Sentence boundaries followed the corrected segmentation of the stimuli (305 sentences). Figure 5 shows the surprisal estimates from the three contexts for one sentence.

In the second-stage models, the variance of one random intercept was estimated at zero (singular fit) for the full–sentential comparison in FFD and for the full–sentential comparisons in RRD, as well as for some of the n-gram comparisons (GD: 7, 10, 100 and 200 words; TD: 100 words). All first-stage models converged without warnings or boundary estimates. The test of a zero mean difference used a null model with no fixed effects and the same random effects, fitted through the modular `lme4` interface, since the formula interface adds an intercept. LRTs used maximum likelihood, and intervals and Satterthwaite tests used REML refits (`lmerTest`; Kuznetsova et al., 2017).

C Memorized Passage

Figure 6 shows the greedy continuations of the six LMs after the first 25 words of the Human Rights passage. Kanarya-2b reproduced the next 36 words exactly and 110 words with one error. Kanarya-750m reproduced the first three words and then restarted the preamble. The other models produced text on the same topic but not the original wording.

D Additional Results

Table 6 gives the estimates plotted in Figure 1. Figure 7 shows the sentential–2-gram and full–sentential comparisons, Table 7 the model-size comparisons summarized in §4.3, and Table 8 the comprehension slope for each LM. Table 9 gives the genre and question-type slopes for TD, which did not differ significantly. For GD, the corresponding slopes lay between -0.01 and 0.16 and none was significant. Table 10 gives the sentential–2-gram and full–sentential estimates for each LM (nine passages).

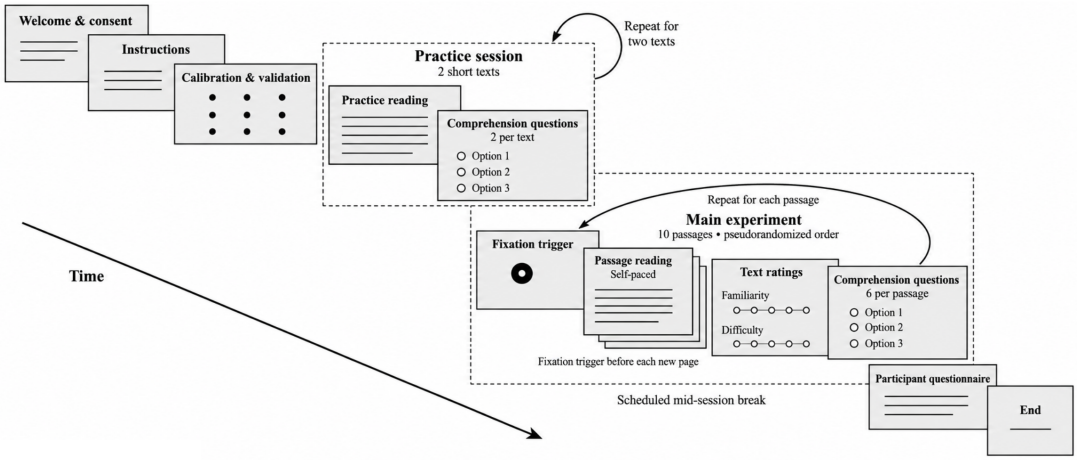


Figure 4: Procedure of a session. Participants read the 10 passages in pseudo-randomized order, each followed by ratings and six comprehension questions; a fixation trigger preceded each screen.

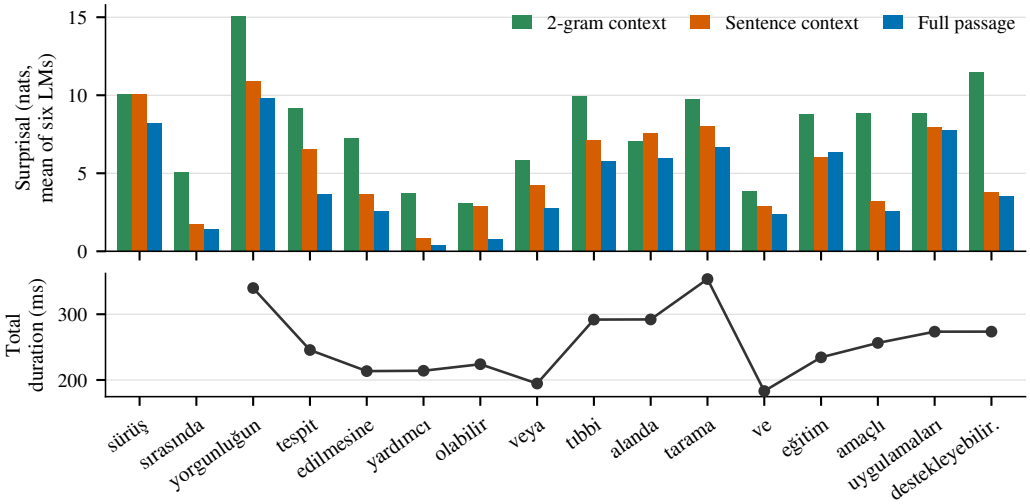


Figure 5: Surprisal under the three contexts (mean of six LMs) and mean total duration for one sentence of the MultiplEYE passage ('For example, it can help detect fatigue while driving, or support screening and training applications in medicine'). The sentence-initial word is omitted; total duration starts at the fourth word, as in the analyses (§3.3).

Model (run / block)	Continuation
Prompt	İnsan Hakları Evrensel Beyannamesi - Önsöz İnsanlık ailesinin bütün üyelerinde bulunan haysiyetin ve bunların eşit ve devir kabul etmez haklarının tanınması hususunun, hürriyetin, adaletin ve
Original	dünya barışının temeli olmasına, İnsan haklarının tanınmaması ve hor görülmesinin insanlık vicdanını isyana sevkeden vahşiliklere sebep olmuş bulunmasına, dehşetten ve yoksulluktan kurtulmuş insanların, içinde söz ve inanma hürriyetlerine sahip olacakları bir dünyanın kurulması en yüksek amaçları olarak ilan edilmiş bulunmasına,
Kanarya-2b (36 / 110)	dünya barışının temeli olmasına, İnsan haklarının tanınmaması ve hor görülmesinin insanlık vicdanını isyana sevkeden vahşiliklere sebep olmuş bulunmasına, dehşetten ve yoksulluktan kurtulmuş insanların, içinde söz ve inanma hürriyetlerine sahip olacakları bir dünyanın kurulması en yüksek amaçları oralak ilan edilmiş bulunmasına,
Kanarya-750m (3 / 3)	dünya barışının temeli olarak ilan etmiştir. İnsan Hakları Evrensel Beyannamesi - Önsöz İnsanlık ailesinin bütün üyelerinde bulunan haysiyetin ve bunların eşit ve devir kabul etmez haklarının tanınması hususunun, hürriyetin, adaletin ve dünya barışının temeli olarak ilan etmiştir. İnsan Hakları Evrensel
cosmosGPT-L (0 / 2)	barışın temeli olduğu, bu hakların ancak bütün insanların doğuştan sahip oldukları ve eşit olarak yararlanabilecekleri haklardan ibaret bulunduğu, bu hakların ancak bütün insanların ortak çabalarıyla ve ortak çabalarıyla korunabileceği, bu hakların ancak bütün insanların ortak çabalarıyla ve ortak çabalarıyla korunabileceği,
cosmosGPT-M (0 / 3)	hukukun üstünlüğünün, insan haklarının ve temel hürriyetlerin en önemli teminatı olduğunu kabul eden ve bu hakların korunması için gerekli tedbirleri almayı ödev bilen bir insan hakları bildirgesidir. İnsan Hakları Evrensel Beyannamesi, Birleşmiş Milletler Genel Kurulu tarafından, 1 Aralık 1948 tarihinde
cosmosGPT-S (0 / 1)	barışın sağlanması için gerekli olduğu, bu hakların, insan hakları ve demokrasi ile bağdaşmadığı, bu hakların, insan hakları ve demokrasi ile bağdaşmadığı, bu hakların, insan hakları ve demokrasi ile bağdaşmadığı, bu hakların, insan hakları ve demokrasi ile bağdaşmadığı, bu hakların, insan
mGPT (0 / 2)	eşitliğin temel ilkelerinden biri olduğu, bu ilkelerin insan hakları temelinde korunması ve geliştirilmesi gerektiği, insan haklarının korunması ve geliştirilmesi için gerekli olan her türlü tedbiri almak ve uygulamakla yükümlü olduğu, bu amaçla, insan hakları evrensel beyannamesi, insan hakları konusunda uluslararası

Figure 6: Greedy continuations (first 40 words) after the first 25 words of the Human Rights passage. Highlighted: exact matches of at least two words. In parentheses: exact words directly after this prompt / longest exact block.

	GD		TD	
Context (n-gram)	Full − context	p_H	Full − context	p_H
2 words	2.11 [0.72, 3.49]	.039	4.14 [2.32, 5.95]	.002
3 words	1.64 [0.32, 2.95]	.111	3.19 [1.53, 4.86]	.008
5 words	1.15 [−0.30, 2.61]	.612	2.31 [0.34, 4.29]	.128
7 words	0.52 [−1.02, 2.06]	1.000	1.33 [−0.64, 3.30]	.740
10 words	0.59 [−0.78, 1.96]	1.000	1.30 [−0.63, 3.23]	.740
20 words	0.15 [−0.95, 1.26]	1.000	0.70 [−0.70, 2.09]	.744
50 words	−0.13 [−1.08, 0.83]	1.000	0.12 [−0.85, 1.10]	.876
100 words	−0.03 [−0.67, 0.60]	1.000	0.22 [−0.45, 0.88]	.876
200 words	0.21 [−0.08, 0.50]	.735	0.33 [−0.20, 0.87]	.740
Sentential	0.35 [−0.84, 1.54]	1.000	1.64 [0.05, 3.23]	.206

Table 6: Full context minus each n-gram context and the sentential context, as plotted in Figure 1 (mean of six LMs, nine passages; ΔLL per 1,000 observations with 95% Satterthwaite intervals). p_H: LRT, Holm-corrected over the ten contexts per measure.

Contrast	Measure	Diff. [CI]	p_H	$p_{S,H}$
Kanarya: 750m vs. 2b				
Full	GD	0.43 [−0.20, 1.06]	.478	.617
Full	TD	0.55 [−0.35, 1.46]	.478	.617
2-gram loss	GD	0.26 [−0.31, 0.83]	.405	.509
2-gram loss	TD	0.72 [−0.32, 1.76]	.405	.509
cosmosGPT: mean of S and M vs. L				
Full	GD	0.17 [−0.14, 0.49]	.478	.617
Full	TD	0.19 [−0.13, 0.51]	.478	.617
2-gram loss	GD	0.28 [−0.09, 0.65]	.405	.509
2-gram loss	TD	0.28 [−0.17, 0.74]	.405	.509

Table 7: Model size within families, without the Human Rights passage. Full: ΔLL of the smaller minus the larger model(s) with full context. 2-gram loss: fit lost from full context to the 2-gram context by the smaller minus the larger model(s); positive values mean that the larger model lost less. Holm correction over four comparisons per contrast.

LM	All 10 passages	Without HR
Six-LM mean	0.39 [0.08, 0.70]	0.44 [0.11, 0.77]
Kanarya-750m	0.43 [0.08, 0.78]	0.50 [0.12, 0.87]
Kanarya-2b	0.44 [0.07, 0.82]	0.51 [0.11, 0.91]
cosmosGPT-S	0.33 [0.05, 0.61]	0.36 [0.06, 0.67]
cosmosGPT-M	0.41 [0.08, 0.75]	0.46 [0.10, 0.83]
cosmosGPT-L	0.42 [0.10, 0.74]	0.46 [0.12, 0.81]
mGPT	0.31 [0.03, 0.58]	0.34 [0.05, 0.63]

Table 8: Accuracy slope (per SD) for the benefit of full over sentential context in TD, for each LM and their mean, with 95% Satterthwaite intervals; reading speed controlled.

Accuracy slope (per SD), TD	All 10 passages	Without HR
By genre		
Argumentative	0.60 [−0.12, 1.32]	0.60 [−0.22, 1.41]
Institutional	0.06 [−0.66, 0.78]	0.19 [−0.97, 1.34]
Literary	0.37 [−0.14, 0.88]	0.37 [−0.21, 0.95]
Popular science	0.54 [−0.18, 1.26]	0.54 [−0.28, 1.36]
Accuracy × genre	$\chi^2(3) = 2.94, p = .401$	$\chi^2(3) = 1.25, p = .741$
By question type (separate models)		
Local	0.23 [−0.07, 0.53]	0.25 [−0.09, 0.59]
Bridging	0.31 [0.08, 0.55]	0.37 [0.10, 0.64]
Global	0.29 [0.01, 0.57]	0.32 [0.01, 0.63]
Equal slopes (joint model)	$\chi^2(2) = 1.45, p = .485$	$\chi^2(2) = 1.66, p = .437$

Table 9: Accuracy slopes (per SD) for TD by genre and by question type, with 95% Satterthwaite intervals. Genre slopes come from one model with genre-specific slopes; question-type slopes come from separate models. The tests compare these models with models without the interaction or with equal slopes.

LM	FFD	GD	TD	RRD
Sentential − 2-gram				
Six-LM mean	0.90	1.76	2.49	0.96
Kanarya-750m	0.84	1.66	2.45	1.02
Kanarya-2b	0.72	1.33	1.74	0.87
cosmosGPT-S	0.88	1.83	2.74	1.09
cosmosGPT-M	0.85	1.63	2.40	0.98
cosmosGPT-L	0.92	1.64	2.47	0.80
mGPT	1.17	2.45	3.18	0.98
Full − sentential				
Six-LM mean	0.43	0.35	1.64	0.24
Kanarya-750m	0.50	0.42	1.85	0.52
Kanarya-2b	0.52	0.49	1.83	0.24
cosmosGPT-S	0.49	0.40	1.56	0.12
cosmosGPT-M	0.34	0.21	1.50	0.10
cosmosGPT-L	0.25	0.12	1.34	0.12
mGPT	0.50	0.45	1.77	0.37

Table 10: Sentential − 2-gram and full − sentential differences for each LM and their mean (nine passages; ΔLL per 1,000 observations).

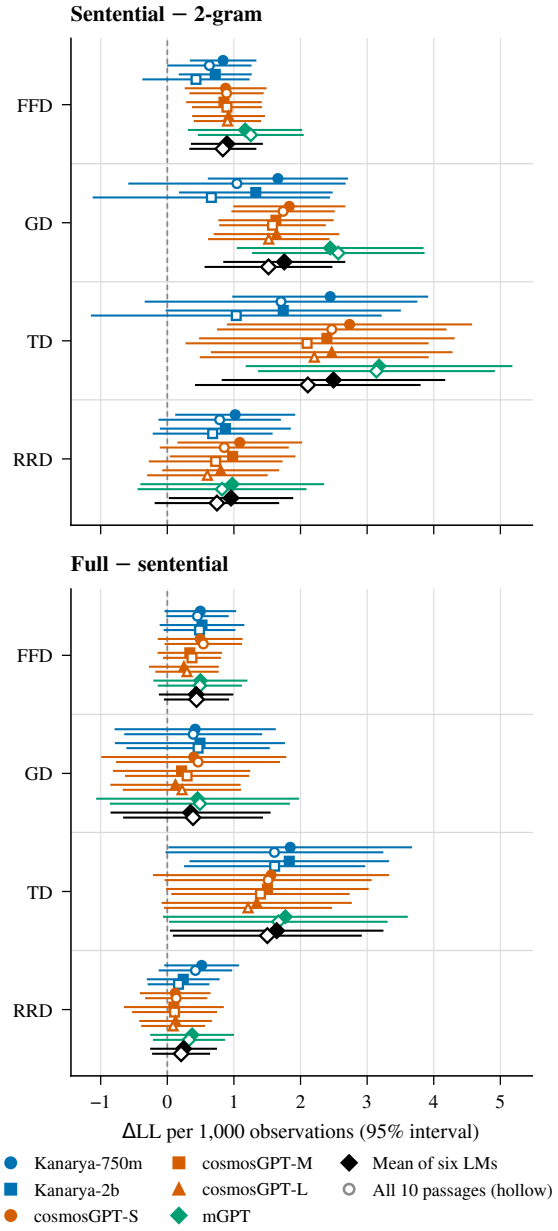


Figure 7: Sentential − 2-gram and full − sentential comparisons for each LM and for their mean (ΔLL per 1,000 observations with 95% Satterthwaite intervals), with the nine passages of the main analysis (filled) and with all 10 passages (hollow).